

E E 385V: Brain Computer Interaction Due Date: 30th September 2026 Homework #1

Course Policy: Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- Answer to all below questions with requested figures.
- Assignments are due at the end of day.
- Report is individual for each student.
- Group work is encouraged.
- Reports delivered via Canvas.
- Be concise

Qs 1: Spatial filter on Motor Imagery data

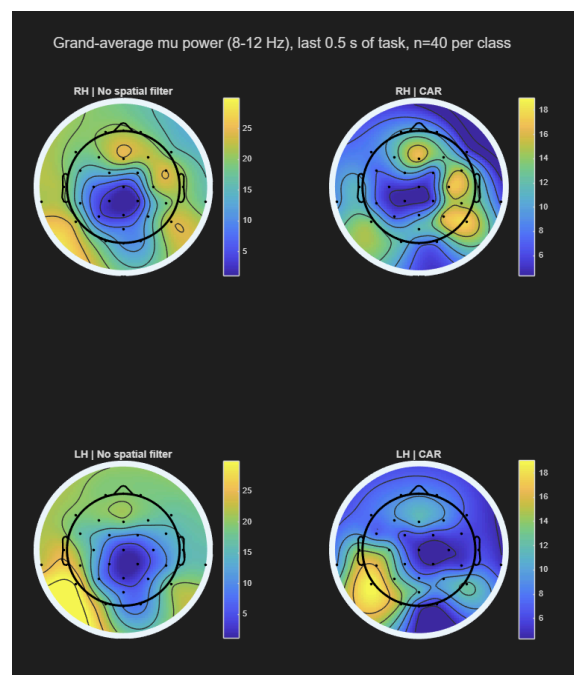
- What is motor imagery? What are the neurophysiological correlates of motor imagery?
 - Motor imagery is the mental playing of a movement. Examples are imagining holding an object or moving your arm without actually doing so in real life.
 - Correlates are changes in sensorimotor rhythms (SMRs).
 - Event related desynchronization: decrease in sync within a neural network, seen as attenuation of amplitude
 - Event related synchronization: increase in sync
 - Mu band: The slides use 8 to 11 Hz in the bandpass example and 9 to 13 Hz in the topographies. Power drops (ERD) over the sensorimotor cortex during imagery.
 - Beta band: The slides show a beta example at 26 to 30 Hz and a foot-imagery ERS at 18 to 23 Hz. Beta also shows a rebound (ERS) after the movement or imagery ends.
 - Lateralization: Left-hand and right-hand imagery produce ERD over the opposite (contralateral) hemisphere. The slides show this as different mu ERD topographies for the two hands. This is why your HW compares RH and LH topoplots.
 - Somatotopy: Foot imagery gives a different spatial pattern from hand imagery.
- Load the EEG data from the session (1 session in total) and run. Apply temporal filtering in the mu band and a spatial filtering (CAR or Laplacian) of your choice.
 - Describe why we temporally and spatially filter EEG data.
 - Temporal (mu band, 8-12 Hz): Raw EEG is dominated by drift, line noise, EMG and other rhythms. A band-pass isolates the sensorimotor mu rhythm, which drops (ERD) over the sensorimotor cortex during motor imagery. Channel FP1 had a mu-band power fraction of about 0.02-0.03 before filtering and 0.968-0.979 after, in all four runs. The filter was stable, with no NaN or Inf values.
 - Spatial (CAR): Scalp EEG is smeared by volume conduction and depends on the reference. CAR subtracts the average of all channels from each channel. This removes noise common to all channels and sharpens the topography. The Laplacian is the alternative. It subtracts a neighbor average, emphasizes local sources, and is reference independent.
- Extract task trials for each class (RH and LH) based on the trigger information provided.
 - Can we deduce the number of trials? If so, how many?
 - Yes. There are 40 RH and 40 LH trials, 80 in total. Each of the 4 runs has 10 events of code 769 and 10 of code 770, which gives 10 RH and 10 LH per run. The 20 events each of 768 and 1000 are consistent with 20 trials per run. The

codes 7691-7703 look like hit/miss or feedback markers. That fits the code's tally of RH 32 hit and 8 miss, and LH 25 hit and 15 miss. Trial lengths ranged from 1.24 to 7.06 s for RH and 2.10 to 7.06 s for LH.

- Compute the mu power of each task trial (s^2)
 - Per-trial mu power was computed as the mean of the squared band-passed signal over each trial. The result is a 32 channel $\times$ 40 trial matrix for each of RH raw, RH CAR, LH raw and LH CAR.

Condition	Grand-average range
RH, no spatial filter	2.14 to 24.3
RH, CAR	4.32 to 17.6
LH, no spatial filter	1.06 to 29.9
LH, CAR	4.49 to 19

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- Show the topological plot of the grand averaged mu power from the last 0.5s of the task trials, with and without spatial filtering. Do these for each class.
 - Describe your observations.
 - No spatial filter: For both classes the lowest mu power is a single blob over central and centroparietal sites (the Cz/CP area). Mu is highest at the posterior-left and posterior edges, and RH also shows a few patches of higher power over frontal and temporal sites. The two classes look similar and show no clear lateralization. This is the broad, shared pattern you would expect from volume conduction and reference effects.
 - CAR, RH: The minimum shifts to the left central region (near C3/CP1), the contralateral side. Power is higher over right temporal and parietal sites, ipsilateral to the imagined hand. This is the expected contralateral ERD.
 - CAR, LH: The minimum sits at the center and slightly to the right (around Cz/C4/CP2). Posterior-left parietal and occipital power is high. This is a weaker contralateral ERD.
 - Scale change: The color range shrinks from about 2-30 to about 4-19 after CAR. CAR removes the common-mode power and sharpens spatial contrast.



Trial structure and trigger:

All the data files provided are stored in general data format (GDF). The .gdf files can be loaded with the function provided in MATLAB as follows.

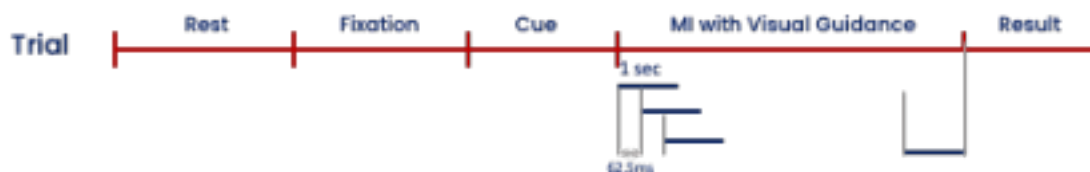
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[s,h]=sload('fileName.gdf');
```

The above command will result in two variables in the MATLAB workspace. The variable s(Signal variable) will contain the Signals. The signal variable contains 36 columns (first 32 are EEG channels, and the last 4 columns should be neglected for analysis). Variable h(header structure) will hold the detailed information of various parameters involved in conducting the experiment. The variable h includes the event information that describes the structure of EEG data recorded over time. The following fields describe event structure overtime for this dataset:

- h.EVENT.TYP : This variable provides the trigger value corresponding to an event in the data
- h.EVENT.POS : This variable provides the position of the corresponding trigger on the sample scale.

Trigger	Description
1000	Trial start
768	Fixation
769/770	Cue (RH/LH)
7691/7701	Continuous feedback starts (the bar starts moving)(RH/LH)
7692/7702	Target misses or time-out
7693/7703	Target hit (the accumulated probabilities reached the threshold of the correct class) (RH/LH)

Trial structure:



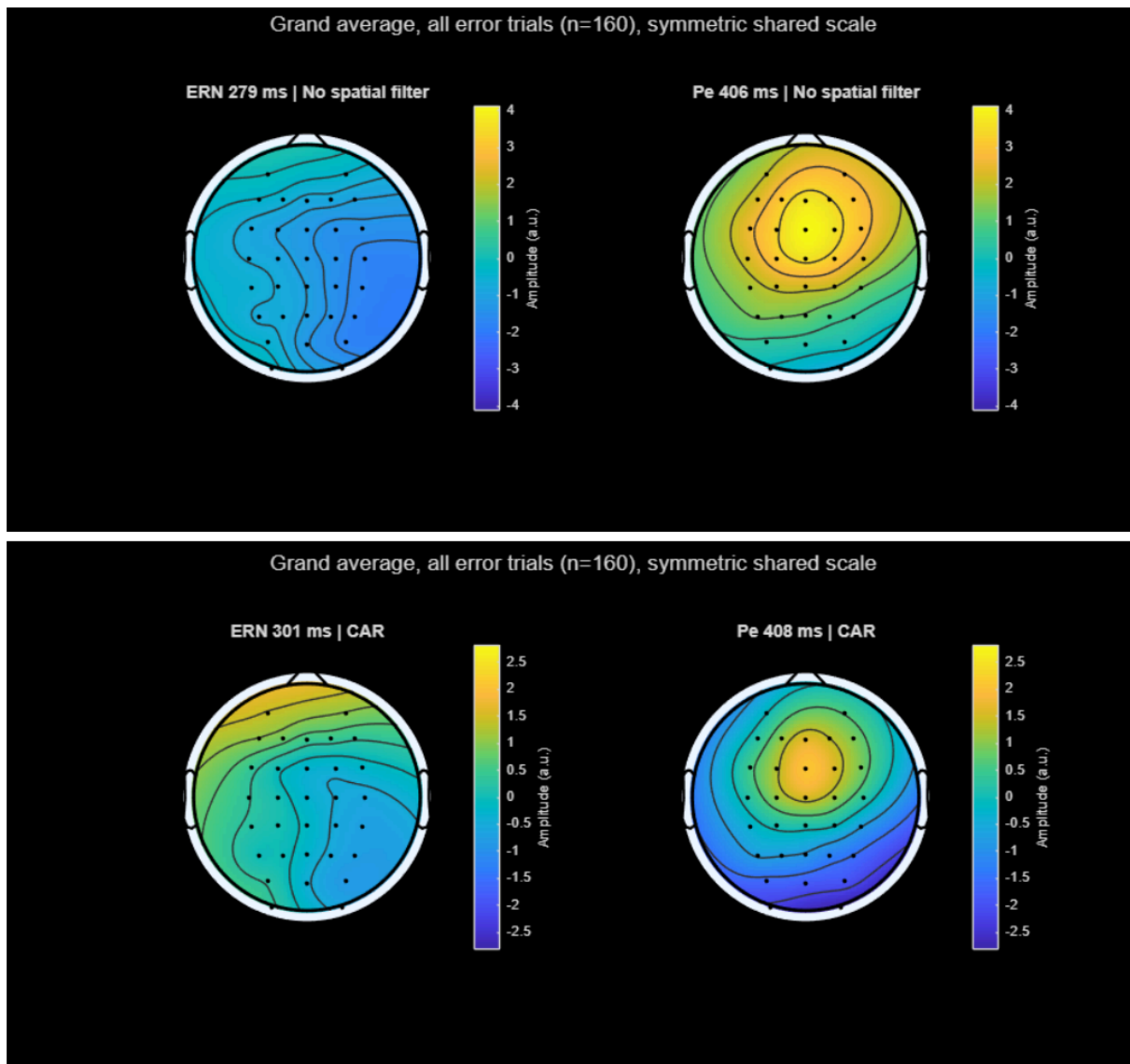
- a sample/epoch of the task period is a one-second window
- samples are generated @16 Hz; That is, with a hop of 62.5ms

Qs 2: Spatial filter on error-related potentials (ErrPs) data

- What are error-related potentials (ErrPs)? What are the neurophysiological correlates of ErrPs?
 - Error-related potentials are event-related potentials evoked when a person perceives an error, whether their own or one made by a system they are controlling. They are time-locked to the error event and are strongest over fronto-central scalp sites. The classic components are an early negative deflection (ERN or Ne) followed by a later positive deflection (Pe). Their

generators are generally attributed to the anterior cingulate and medial frontal cortex.

- ERN (error-related negativity): a fronto-central negative deflection, generated mainly in the anterior cingulate cortex / medial frontal cortex. It reflects error detection and performance monitoring.
- Pe (error positivity): a later, broader positive deflection over central/centro-parietal sites. It is linked to conscious error awareness and evaluation.
- Theta-band (4-8 Hz) activity: error-related theta increase over medial frontal sites, which is why the data were filtered in this band.
- Description of the ErrP data provided:
 - **rotation_data**: 320 trials from 32 channels, and each trial is of length 2s; [-1s 1s] with respect to the trigger onset
 - **label**: 0 marks a correct trial and 1 marks an error trial
 - **magnitude**: magnitudes can be of 0, 3, 6, 9, 12 degrees; the larger the magnitude, the more obvious the erroneous event.
 - **fileID**: run number
 - **sessionID**: session number
 - The ErrP has already been filtered in the theta band.
- Plot the grand average of the ErrP data for each magnitude, use data [-0.2s 0.8s] w.r.t trigger onset on Cz.
 - Trial counts: 160 correct (0°) and 40 each at 3°, 6°, 9° and 12°. Larger rotations produce a clearly larger late positivity. The 12° curve peaks at about 8.3 a.u. near 390 ms, while the 0° (correct) curve stays flat near 0. Smaller magnitudes show weaker, less consistent responses.
- Identify the ERN (first negative peak) and Pe (first positive peak) time points. • Plot the topological plot of the grand averaged amplitude at the Pe time point, using all the error trials. Do the same for ERN time point.
 - Describe your findings.
 - Latencies (all-error grand average at Cz, no spatial filter):
 - ERN: 279 ms (-1.20 a.u.)
 - Pe: 406 ms (+3.62 a.u.)
 - Topoplots (raw):
 - ERN (279 ms): broadly negative across the scalp (spatial mean -1.01), weakest at the front and most negative posterior/right. It is diffuse, with no sharp fronto-central focus.
 - Pe (406 ms): a clear positive central/fronto-central focus (spatial mean +2.06), decaying toward the periphery. This matches the expected Pe topography.
 - Findings: The Pe is clean and well localized. The ERN is small and diffuse, partly because it is a weak component in the averaged Cz trace and the raw data share a common offset across channels. Its timing (~280 ms) is later than a classic ERN, but that is plausible for a visually-triggered, perceptual error.



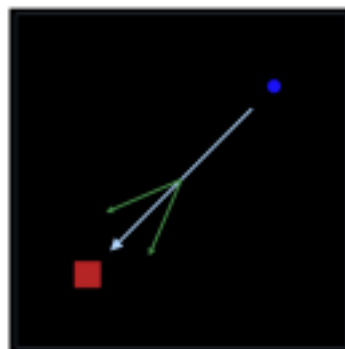
- Perform CAR on the data, and plot the grand average of the ErrP data at Cz for each magnitude; use data [-0.2s 0.8s] w.r.t trigger onset.
 - Visualise the topological plot at Pe and ERN time points.

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Component	No filter	CAR	Shift
ERN	279 ms	301 ms	+21 ms
Pe	406 ms	408 ms	+2 ms

- Topoplots after CAR:
 - ERN (301 ms): the spatial mean is zero by construction. Frontal/left-frontal sites are positive and the rest of the scalp is near zero to slightly negative.
 - Pe (408 ms): a focal central positivity (Cz/FCz area) with negative values at the periphery, especially posterior and lateral sites. The focus is sharper than in the raw map.
- Identify one potential disadvantage of CAR.

- It forces the channel mean to zero, so activity from a strong source is spread into all other channels as an opposite-sign artifact. This distorts the true topography and can shift peak latencies and amplitudes. Here, the ERN at Cz dropped to -0.30 a.u. and moved 21 ms, and the 12° ERN even turned slightly positive. CAR also assumes uniform electrode coverage, and with only 32 channels and no inferior coverage, that is violated.
- Use CCA to build spatial filters on the data provided. Plot the topological plots of the weights of the first 5 CCA components (one topo plot for each), describe your findings. Note, there are two classes: error class and correct class.
 - Correlations are modest and decrease steadily, so no component dominates. They are also optimistic, because the filters were fit and evaluated on the same trials with no held-out test.
 - CCA 1 ($r = 0.285$): strongest weights are a left/posterior-temporal negative lobe and positive right-central/parietal lobes. It is a spatially structured contrast, not a simple Cz-centered pattern.
 - CCA 2-5: progressively more fragmented patterns, with alternating-sign weights at several sites (multiple small positive/negative foci). These capture additional, weaker error-vs-correct differences or noise.
 - Interpretation caveat: the weights are a backward model (filter weights), not forward activation patterns, so they reflect noise suppression as well as signal location. Alternating polarity across neighboring electrodes is typical of such filters cancelling shared noise, and the weights should not be read as source locations.
 - Compared with CAR: the CCA filters are data-driven and class-specific, whereas CAR is a fixed, class-blind operation, so CCA can improve discriminability while CAR only removes common-mode signal.



Description of the ErrP task: Participants controlled a cursor from start (top-right) to goal locations (bottom-left) with a joystick as quick and straight as possible, using their left thumbs. In each trial, the normal joystick-cursor mapping can be violated by a rotation magnitude when the cursor exceeded a pre-defined fixed, invisible boundary, which was set to 50% of the diagonal distance between the start and goal locations. Participants were instructed to attempt to perceive the visuo-motor rotation and move the cursor. To control the levels of difficulty to perceive errors, we used four magnitudes of rotation: 3, 6, 9 and 12 degrees. The green arrows indicate directions where the cursor could be rotated.

Reference to learn more about CCA in EEG spatial filtering: Spüler, M., Walter, A., Rosenstiel, W., & Bogdan, M. (2013). Spatial filtering based on canonical correlation analysis for classification of evoked or event-related potentials in EEG data. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 22(6), 1097-1103.